

Direct and Inverse Proportion

Choose the Appropriate Answer

1. Find the ratio of the number of circles to number of squares.



- (a) 2: 4 (b) 2: 3 (c) 4 : 5 (d) 3: 4

Ans: (b) 2: 3

2. Find the ratio 555 g to 5 kg.

- (a) 111: 1000 (b) 111: 1 (c) 111: 100 (d) 555: 5

Ans: (a) 111: 1000

$$\begin{aligned} 1 \text{ kg} &= 1000 \text{ g}, \quad 5 \text{ kg} = 5000 \text{ g} \\ 555 \text{ g} : 5 \text{ kg} &= 555 \text{ g} : 5000 \text{ g} = 111 : 1000 \end{aligned}$$

3. Find the ratio 21 km to 175 m.

- (a) 1200: 1 (b) 120: 1 (c) 21: 175 (d) 21: 1750

Ans: (b) 120: 1

$$\begin{aligned} 1 \text{ km} &= 1000 \text{ m}, \quad 21 \text{ km} = 21 \times 1000 = 21,000 \text{ m} \\ \therefore 21 \text{ km} : 175 \text{ m} &= 21000 : 175 = 120 : 1 \end{aligned}$$

4. To find the equation of Direct Proportion

- (a) $x y = \text{Constant}$ (b) $x / y = \text{Constant}$ (c) both a & b are wrong (d) None

Ans: (b) $x / y = \text{Constant}$

5. To find the equation of Inverse Proportion

- (a) $x y = \text{Constant}$ (b) $x / y = \text{Constant}$ (c) both a & b are wrong (d) None

Ans: (a) $x y = \text{Constant}$

6. If the cost of 3 books is Rs.90, then find the cost of 12 books.

- (a) Rs.300 (b) Rs.320 (c) Rs.360 (d) Rs.400

Ans: (c) Rs.360

$$\frac{3}{90} = \frac{12}{x} \Rightarrow x = \frac{12 \times 90}{3} = 360$$

7. If Mani buys 5kg of potatoes for Rs.75 then he can buy ___kg of potatoes for Rs.105.

- (a) 6 (b) 7 (c) 8 (d) 5

Ans: (b) 7

$$\frac{5}{75} = \frac{x}{105} \Rightarrow x = \frac{5 \times 105}{75} = 7$$

8. 35 cycles were produced in 5 days by a company then _____ cycles will be produced in 21 days.

(a) 150 (b) 70 (c) 100 (d) 147

Ans: (d) 147

$$\frac{35}{5} = \frac{x}{21} \Rightarrow x = \frac{\cancel{35}^7 \times 21}{\cancel{5}_1} = 147$$

9. An aircraft can accommodate 280 people in 2 trips. It can take _____ trips to take 1400 people.

(a) 8 (b) 10 (c) 9 (d) 12

Ans: (b) 10

10. Suppose 3 Kg. of sugar is used to prepare sweets for 50 members, then _____ Kg. of Sugar is required for 150 members.

(a) 9 (b) 10 (c) 15 (d) 6

Ans: (a) 9

$$\frac{3}{50} = \frac{x}{150} \Rightarrow x = \frac{3 \times \cancel{150}^3}{\cancel{50}_1} = 9$$

11. If the cost of 8 apples is 56 then the cost of 12 apples is _____.

(a) 54 (b) 84 (c) 66 (d) 72

Ans: (b) 84

12. If the weight of one fruit box is 3.5 Kg. Then the weight of 6 such boxes is _____.

(a) 21 Kg (b) 32 Kg (c) 18 Kg (d) 24 Kg

Ans: (a) 21 Kg

13. A car travels 60 km with 3 liters of petrol. If the car has to cover the distance of 200 Km, it requires _____ liters of petrol.

(a) 20 (b) 6 (c) 10 (d) 8

Ans: (c) 10

14. If the cost of 7 m cloth is Rs.294, then the cost of 5 m of cloth is _____.

(a) Rs.200 (b) Rs.210 (c) Rs.220 (d) Rs.230

Ans: (b) Rs.210

15. If a machine in a cool drinks factory fills 600 bottles in 5 hrs. Then it will fill _____ bottles in 3 hours.

(a) 310 (b) 330 (c) 350 (d) 360

Ans: (d) 360

16. Distance travelled by a bus and time taken are in direct proportion.

- (a) True (b) False (c) Inverse Proportion (d) None

Ans: (a) True

17. Expenditure of a family to number of members of the family are in ----- proportion.

- (a) Inverse (b) Direct (c) Joint (d) None

Ans: (b) Direct

18. Number of students in a hostel and consumption of food are ----- proportion.

- (a) Inverse (b) Joint (c) Direct (d) None

Ans: (c) Direct

19. 16 taps can fill a petrol tank in 18 minutes. The time taken for 9 taps to fill the same tank will be _____ minutes.

- (a) 28 (b) 34 (c) 32 (d) 30

Ans: (c) 32

$$\begin{aligned} 16 \times 18 &= 9 \times X \\ X &= \frac{16 \times 18}{9} = 32 \end{aligned}$$

20. If 40 workers can do a project work in 8 days, then _____ workers can do it in 4 days.

- (a) 60 (b) 80 (c) 70 (d) 50

Ans: (b) 80

21. 12 cows can graze a field for 10 days. 20 cows can graze the same field for _____ days.

- (a) 15 (b) 18 (c) 6 (d) 8

Ans: (c) 6

$$12 \times 10 = 20 \times x; \quad x = \frac{12 \times 10}{20} = 6$$

22. 4 typists are employed to complete a work in 12 days. If two more typists are added, they will finish the same work in _____ days.

- (a) 7 (b) 8 (c) 9 (d) 10

Ans: (b) 8

$$4 \times 12 = 6 \times x; \quad \frac{4 \times \cancel{12}^2}{\cancel{6}} = x, \quad x = 8$$

23. If Mani buys 5 kg of potatoes for ₹ 75 then he can buy ₹ 105.

- (a) 6 (b) 7 (c) 8 (d) 5

Ans: (b) 7

24. 4 typists are employed to complete a work in 12 days. If two more typists are added, they will finish the same work in days.

- (a) 7 (b) 8 (c) 9 (d) 10

Ans: (b) 8

25. If the cost of 3 books is ₹ 90, then find the cost of 12 books.

- (a) ₹ 300 (b) ₹ 320 (c) ₹ 360 (d) ₹ 400

Ans: (c) 360

26. Suppose 3 kg of sugar is used to prepare sweets for 50 members, then ____ kg of sugar is required for 150 members.

- (a) 9 (b) 10 (c) 15 (d) 6

Ans: (a) 9

27. 12 cows can graze a field for 10 days. 20 cows can graze the same field for ____ days.

- (a) 15 (b) 18 (c) 6 (d) 8

Ans: (c) 6

28. 35 cycles were produced in 5 days by a company then ____ cycles will be produced in 21 days.

- (a) 150 (b) 70 (c) 100 (d) 147

Ans: (d) 147

29. An aircraft can accommodate 280 people in 2 trips. It can take _____ trips to take 1400 people.

- (a) 8 (b) 10 (c) 9 (d) 12

Ans: (b) 10

30. The ratio of 50 paise to Rs.1 is

- (a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 1 : 5

Ans: (a) 1 : 2

$$\text{Rs } 1 = 100 \text{ paise} \Rightarrow 50 \text{ paise} : 100 \text{ paise or } 50 : 100 \text{ or } 1 : 2$$

31. The ratio of 10 m to 1 km is

- (a) 1 : 10 (b) 10 : 1 (c) 1 : 100 (d) 100 : 1

Ans: (c) 1 : 100

$$1 \text{ km} = 1000 \text{ m} \Rightarrow 10 \text{ m} : 1000 \text{ m or } 10 : 1000 \text{ or } 1 : 100$$

32. The ratio of Speed of a cycle 15 km per hour to the speed of scooter 30 km per hour is 1 : 2.

- (a) True (b) False (c) Ambiguous (d) Cannot be determined

Ans: (a) True

$$\text{Required Ratio is } = 15/30 = 1/2 = 1 : 2$$

33. There are 50 students in a class of which 40 are boys and the rest are girls. The ratio of the number of boys and number of girls is
(a) 2:3 (b) 1:5 (c) 4:1 (d) 2:5

Ans: (c) 4:1

$$\begin{aligned}\text{Number of Girls} &= 50 - 40 = 10 \\ \text{Required ratio} &= 40 : 10 = 4 : 1\end{aligned}$$

34. The ratio of speed of cycle 12 km / hour to the speed of scooter 36 km / hour is
(a) 1:2 (b) 1:3 (c) 1:4 (d) None of the above

Ans: (b) 1:3

$$\text{Speed of cycle/Speed of scooter} = 12/36 = 1/3 = 1:3$$

35. Number of note books purchased and its cost is the example of ----- proportion.
(a) Inverse (b) Direct (c) Joint (d) None

Ans: (b) Direct

36. Food shared by number of students to the fixed quantity of food is the example of ----- proportion.

(a) Inverse (b) Direct (c) Joint (d) None

Ans: (a) Inverse

37. Find the type of proportion for the below example

No. of chocolates	1	2	3	4	5
Price in rupees (₹)	5	10	15	20	25

(a) Inverse (b) Direct (c) Joint (d) None

Ans: (b) Direct

38. Number of boxes of same size and their weight is the example of ----- proportion.
(a) Inverse (b) Direct (c) Joint (d) None

Ans: (b) Direct

39. Number of uniforms to the number of students is the example of -----proportion.
(a) Inverse (b) Direct (c) Joint (d) None

Ans: (b) Direct

40. Speed of a vehicle and the time taken to cover the fixed distance is the example of ----- proportion.

(a) Inverse (b) Direct (c) Joint (d) None

Ans: (a) Inverse

41. Find the type of proportion for the below example.

No. of workers	1	2	4	5
Time in hours	20	10	5	2.5

- (a) Inverse (b) Direct (c) Joint (d) None

Ans: (a) Inverse

42. When we compare two quantities by division we call it

- (a) Ratio (b) Proportion (c) Decimal (d) Division

Ans: (a) Ratio

43. Ratio of Rs.215 and 230 gram

- (a) 43:46 (b) 40:45 (c) Can't be determined (d) 1:2

Ans: (c) Can't be determined

Cannot directly compare or find the ratio of different units like Rupees (Money) and Grams (Mass)

44. If four numbers are in proportion then the middle terms are called....

- (a) Extremes (b) Means (c) Medians (d) None

Ans: (b) Means

In a proportion of four terms (a:b::c:d), the first and the last terms (a and d) are known as the extremes, while the two middle terms (b and c) are called the means.

45. I travelled 3 Km in 2 hours and my friend travelled 5 Km in 3 hours, the ratio our speed is

- (a) 2:3 (b) 3:5 (c) 9:10 (d) 10:9

Ans: (c) 9:10

$$\text{Speed}_1 = \frac{3 \text{ Km}}{2 \text{ hours}} = \frac{3}{2} \text{ Km/hr.} \quad \text{Speed}_2 = \frac{5 \text{ Km}}{3 \text{ hours}} = \frac{5}{3} \text{ Km/hr.}$$

$$\text{The ratio of their speeds} = \frac{\text{Speed}_1}{\text{Speed}_2} = \frac{\frac{3}{2}}{\frac{5}{3}} = 9 : 10.$$

46. The ratio of 2 hrs. to 300 min is

- (a) 5:2 (b) 2:5 (c) 3:5 (d) 2:3

Ans: (b) 2:5

$$2 \text{ hours} \times 60 \text{ minutes/hour} = 120 \text{ minutes.}$$

$$\text{The ratio is} = \frac{120}{300} = 2 : 5$$

47. If a: b = c: d, they are in

- (a) Proportion (b) Ratio (c) Fraction (d) None

Ans: (a) Proportion

48. In a class of 45 students, girls to boy's ratio is 5:4, then the number of girls in the class is....

- (a) 20 (b) 25 (c) 30 (d) 35

Ans: (b) 25

$$\text{Number of girls} = 45 \times \frac{5}{9} = 25$$

49. Rs.720 is divided among two friends in the ratio 5:4, then how much each gets ---.

- (a) 300, 420 (b) 500, 220 (c) 400, 320 (d) 360, 360

Ans: (c) 400, 320

50. The ratio 3: 4 and 5: 6 is -----

- (a) Equal (b) Unequal (c) Equivalent (d) None

Ans: (b) Unequal

51. Find the ratio of 3 km to 300 m.

- (a) 10: 1 (b) 1: 10 (c) 1: 5 (d) None of these

Ans: (a) 10: 1

$$3 \text{ km} = 3 \times 1000 \text{ m} = 3000 \text{ m}$$

52. A map is given with a scale of 2 cm = 1000 km. What is the actual distance between the two places in Km, if the distance in the map is 2.5 cm?

- (a) 1250 km (b) 1500 km (c) 2500 km (d) None of these

Ans: (a) 1250 km

$$1 \text{ cm} = \frac{1000}{2} \text{ km} = 500 \text{ km} \Rightarrow 2.5 \text{ cm} \times 500 \text{ km/cm} = 1250 \text{ km}$$

53. 6 bowls cost Rs.90. What would be the cost of 10 such bowls?

- (a) Rs.300 (b) Rs.150 (c) Rs.200 (d) Rs.250

Ans: (b) Rs.150

54. The car that I own can go 150 km with 25 litres of petrol. How far can it go with 30 litres of petrol?

- (a) 125 km (b) 160 km (c) 250 km (d) None of these

Ans: (d) None of these

$$\begin{aligned} \text{Distance per litre} &= \frac{150 \text{ km}}{25 \text{ litres}} = 6 \text{ km/litre.} \\ \text{Distance with 30 litres} &= 30 \text{ litres} \times 6 \text{ km/litre} = 180 \text{ km} \end{aligned}$$

55. The ratio of 90 cm to 1.5 m is

- (a) 2 : 5 (b) 3 : 5 (c) 4 : 5 (d) None of these

Ans: (b) 3: 5

$$1.5 \times 100 = 150 \text{ cm} \quad \Rightarrow \quad \frac{90}{150} = 3 : 5$$

56. A picnic is being planned in a school for Class VII. Girls are 60% of the total number of students and are 18 in number. The ratio of the number of girls to the number of boys in the class is

- (a) 2 : 3 (b) 3 : 2 (c) 4 : 5 (d) None of these

Ans: (b) 3 : 2

$$\text{Total students} = \frac{18}{60\%} = \frac{18}{0.60} = 30 \quad \Rightarrow \quad \text{Number of boys} = 30 - 18 = 12$$

$$\text{The ratio} = \frac{18}{12} = 3:2$$

57. Find the ratio of Speed of a cycle 15 km per hour to the speed of scooter 30 km per hour.

- (a) 2 : 1 (b) 1 : 2 (c) 4 : 5 (d) None of these

Ans: (b) 1 : 2

58. Find the ratio of 5 m to 10 km

- (a) 2000 : 1 (b) 1 : 2 (c) 1 : 2000 (d) None of these

Ans: (c) 1 : 2000

59. Find the ratio of 50 Paise to Rs.5

- (a) 10 : 1 (b) 1 : 10 (c) 1 : 5 (d) None of these

Ans: (b) 1 : 10

60. A change in one quantity results a corresponding change in other quantity is called

- (a) Proportion (b) Ratio (c) Fraction (d) None

Ans: (a) Proportion

61. Weight of 8 class 7 math books is 7.2 kg. What is the weight of 20 such books?

- (a) 15 kg (b) 18 Kg (c) 20 kg (d) None of these

Ans: (b) 18 Kg

62. Cost of 15 electric bulb is Rs.825. What is the cost of 10 similar electric bulb?

- (a) Rs.450 (b) Rs.650 (c) Rs.550 (d) None of these

Ans: (c) Rs.550

$$\text{calculate single unit value} = \frac{825}{15} = 55$$

$$\text{cost of 10 bulbs} = 55 \times 10 = 550$$

63. 400 ml cooking oil cost Rs.60. What is the cost of 2.5 litres of cooking oil?

- (a) Rs.375 (b) Rs.350 (c) Rs.325 (d) Rs.300

Ans: (a) Rs.375

$$2.5 \text{ litres} = 2.5 \times 1000 \text{ ml} = 2500 \text{ ml.}$$

$$\text{Cost per ml} = \frac{60 \text{ Rs.}}{400 \text{ ml}} = \frac{3}{20} \text{ Rs./ml.}$$

$$\text{Cost} = 2500 \text{ ml} \times \frac{3}{20} \text{ Rs./ml} = 125 \times 3 \text{ Rs.} = 375 \text{ Rs.}$$

64. A motor bike consumes 4 litres of petrol to cover 260 km. Find the amount of petrol it will need to cover 1040 km.

(a) 15 litres (b) 18 litres (c) 17 litres (d) 16 litres

Ans: (d) 16 litres

$$\text{Consumption per km} = \frac{4 \text{ litres}}{260 \text{ km}} = \frac{1}{65} \text{ litres/km.}$$

$$\text{Petrol needed} = 1040 \text{ km} \times \frac{1}{65} \text{ litres/km} = 16 \text{ litres.}$$

65. A person can survive for 15 days with Rs.900. For how many days he can survive with Rs.3600.

(a) 40 Days (b) 50 Days (c) 60 Days (d) 65 Days

Ans: (c) 60 Days

$$\text{Days of survival} = 3600 \text{ Rs} \times \frac{15 \text{ days}}{900 \text{ Rs}} = 60 \text{ days.}$$

66. If 3 kg sweets costs Rs.450, then find the cost of 5.5 kg sweets.

(a) Rs.825 (b) Rs.725 (c) Rs.625 (d) Rs.925

Ans: (a) Rs.825

$$\text{Cost} = 5.5 \text{ kg} \times \frac{450 \text{ Rs.}}{3 \text{ kg}} = 825 \text{ Rs.}$$

67. In a race, Ravi covers 5 km in 20 minutes. How much distance will he cover in 2 hour?

(a) 15 km (b) 20 km (c) 25 km (d) 30 km

Ans: (d) 30 km

$$\text{Distance covered} = 120 \text{ minutes} \times \frac{5 \text{ km}}{20 \text{ minutes}} = 30 \text{ km.}$$

68. A machine makes 500 marbles in 20 minutes. How many marbles will it make in 2 and half hours?

(a) 2750 (b) 3750 (c) 4000 (d) 4250

Ans: (b) 3750

$$2 \text{ hours \& Half an hour} = 120 \text{ minutes} + 30 \text{ minutes} = 150 \text{ minutes.}$$

$$\text{marbles} = \frac{500 \text{ marbles}}{20 \text{ minutes}} \times 150 \text{ minutes} = 3750 \text{ marbles.}$$

69. A woman earns Rs.7500 in 15 days. How much will she earn in the month of December?

- (a) Rs.14500 (b) Rs.15000 (c) Rs.15500 (d) Rs.16000

Ans: (c) Rs.15500

$$\text{Total earnings in December} = \frac{\text{Rs.7500}}{15 \text{ days}} \times 31 \text{ days} = \text{Rs.15500.}$$

70. Direct proportion means that as one quantity increases, the other quantity -----.

- (a) Decreases (b) Increases (c) Both (a) and (b) (d) None of these

Ans: (b) Increases

71. If $\frac{3}{5}$ of water tank is filled in 2 minutes, then how much more time will be required to fill the rest of the tank?

- (a) 60 Seconds (b) 50 Seconds (c) 70 Seconds (d) 80 Seconds

Ans: (d) 80 Seconds

$$\text{The remaining fraction} = 1 - \frac{3}{5} = \frac{5}{5} - \frac{3}{5} = \frac{2}{5}$$

$$\text{Rest of the tank filled in sec} = \frac{2}{5} \times \frac{5}{3} \times 120 = 80 \text{ seconds}$$

2 minutes = 120 seconds

72. 5 men can finish a work in 40 days. If 3 men join them, then the work will be finished in how many days?

- (a) 20 (b) 25 (c) 30 (d) 40

Ans: (b) 25

$$\text{Time taken} = \frac{200 \text{ man-days}}{8 \text{ men}} = 25 \text{ days.}$$

73. 12 men can finish a work in 10 days. How many more men are needed to complete the work in 6 days?

- (a) 8 (b) 10 (c) 6 (d) 12

Ans: (a) 8

$$\text{To complete the same 120 work units in 6 days} = \frac{120 \text{ work units}}{6 \text{ days}} = 20 \text{ men.}$$

$$\text{The required number of men: } 20 \text{ men} - 12 \text{ men} = 8 \text{ men.}$$

74. A motorbike covers a distance in 2 hours 30 minutes at an average speed of 60 km/h. How much time it will take to cover the same distance at a speed of 75 km/h?

- (a) 1 hour (b) 1.5 hour (c) 2 hours (d) 3 hours

Ans: (c) 2 hours

$$\text{Time is converted to hours: } 2 \text{ hours} + \frac{30}{60} \text{ hours} = 2.5 \text{ hours.}$$

$$\text{Time} = \frac{60 \text{ km/h} \times 2.5 \text{ hours}}{75 \text{ km/h}} = \frac{150 \text{ km}}{75 \text{ km/h}} = 2 \text{ hours.}$$

75. If 15 machines can dig a trench in 10 days, then 1 machine can dig it in _____ days.

- (a) 15 (b) 20 (c) 25 (d) 150

Ans: (d) 150

$$15 \text{ machines} \times 10 \text{ days} = 150 \text{ work units} \Rightarrow \frac{150 \text{ work units}}{1 \text{ machine}} = 150 \text{ days.}$$

76. In a hostel, 160 students had food for 30 days. For how many days is the food sufficient for 20 students?

- (a) 120 (b) 240 (c) 100 (d) 420

Ans: (b) 240

$$160 \text{ students} \times 30 \text{ days} = 4800 \text{ food units} \Rightarrow \frac{4800 \text{ food units}}{20 \text{ students}} = 240 \text{ days.}$$

77. 5 dozen eggs are purchased for Rs.480. What will be the cost of 250 eggs?

- (a) Rs.2050 (b) Rs.2000 (c) Rs.1925 (d) Rs.1988

Ans: (b) Rs.2000

$$5 \text{ dozen} \times 12 \text{ eggs/dozen} = 60 \text{ eggs} \Rightarrow \frac{480}{60} \times 250 = 2000$$

78. A machine makes a toy every $1/5$ hour. If it works for $5 \frac{2}{5}$ hours, then how many toys will he make?

- (a) 22 (b) 25 (c) 28 (d) 27

Ans: (d) 27

working time to a fraction: $5 \frac{2}{5} \text{ hours} = \frac{27}{5} \text{ hours.}$

number of toys: $(1 / (1/5) = 5).$

Multiply the terms = $5 \text{ toys/hour} \times \frac{27}{5} \text{ hours} = 27 \text{ toys.}$

79. If 21 girls eat as much as 15 boys, how many girls will eat as much as 35 boys?

- (a) 45 (b) 49 (c) 65 (d) 50

Ans: (b) 49

$$\text{ratio of girls to boys: } 21 \text{ girls} = 15 \text{ boys} \Rightarrow \frac{21}{15} \text{ girls} = 1 \text{ boy}$$
$$\text{Calculate for 35 boys: } (\frac{21}{15}) \times 35 = 49$$

80. If 10 pipes having same diameter can fill a water tank in 24 minutes, then how long 8 pipes would take to fill the tank?

- (a) 30 minutes (b) 35 minutes (c) 32 minutes (d) 40 minutes

Ans: 30 minutes

$$N_1 \times T_1 = N_2 \times T_2 \Rightarrow 10 \times 24 = 8 \times T_2$$

$$T_2 = \frac{10 \times 24}{8} = \frac{240}{8} = 30 \text{ minutes.}$$

81. If 12 meters of a steel pipe weighs 24 kg, then what would be the weight of 18 meters pipe?

- (a) 24 (b) 36 (c) 30 (d) 40

Ans: (b) 36

$$\text{weight per meter} = \frac{24 \text{ kg}}{12 \text{ meters}} = 2 \text{ kg / meter}$$

$$\text{weight of an 18 meters pipe} = 2 \text{ kg / meter} \times 18 \text{ meter} = 36 \text{ kg}$$

82. Inverse proportion means that as one quantity increases, the other quantity -----.

- (a) Decreases (b) Increases (c) Both (a)&(b) (d) None of these

Ans: (a) Decreases

83. In unitary method, to solve a direct proportion problem

- (a) First multiplication then division (b) First division then multiplication
(c) Both (a) & (b) are correct (d) None of these

Ans: (b) First division then multiplication

84. In unitary method, to solve an inverse proportion problem

- (a) First multiplication then division (b) First division then multiplication
(c) Both (a) & (b) are correct (d) None of these

Ans: (a) First multiplication then division

85. If 15 apples cost Rs.270, then what would be the cost of 49 apples?

- (a) Rs.885 (b) Rs.828 (c) Rs.882 (d) None of these

Ans: (c) Rs.882

$$\text{Cost of 1 apple} = \text{Rs. } 270 / 15 = \text{Rs. } 18$$

$$\text{Cost of 49 apples} = \text{Rs. } 18 \times 49 = \text{Rs. } 882$$

86. If 8 kg rice costs Rs.416, then how much rice can be bought for Rs135?

- (a) 25 Kg (b) 26 kg (c) 28 Kg (d) None of these

Ans: (d) None of these

$$\text{Cost of 1 kg rice} = \text{Rs. } 416 / 8 = \text{Rs. } 52$$

$$\text{Quantity of rice for Rs. } 135 = \text{Rs. } 135 / \text{Rs. } 52 \approx 2.596 \text{ kg}$$

87. Travelling 900 km by train costs Rs.280. What would be the fare for a journey of 450 km when a person travels by the same train?

- (a) Rs.245 (b) Rs.250 (c) Rs.240 (d) Rs.140

Ans: (d) Rs.140

$$\text{Fare for 450 km} = \text{Rs. } 280 / 2 = \text{Rs. } 140$$

88. 25 chocolates cost Rs.5.75. What will be the cost of 65 similar chocolates?

- (a) Rs.14.95 (b) Rs.15.95 (c) Rs.14.75 (d) Rs.15.85

Ans: (a) Rs.14.95

$$\begin{aligned} 1 \text{ chocolate cost} &= \frac{5.75}{25} = 0.23 \\ 65 \text{ chocolate cost} &= 65 \times 0.23 = 14.95 \end{aligned}$$

89. If a person travels 576 km in 9 hours, then how much time will he take to cover 1408 km?

- (a) 20 hours (b) 18 hours (c) 22 hours (d) 25 hours

Ans: (c) 22 hours

$$\text{Speed} = \frac{576 \text{ km}}{9 \text{ hours}} = 64 \text{ km/hour} \Rightarrow \text{Time} = \frac{1408 \text{ km}}{64 \text{ km/hour}} = 22 \text{ hours.}$$

90. 1152 packets of biscuit can be packed in 12 cartons of the same size. How many similar cartons would be needed to pack 4608 biscuit packets?

- (a) 84 (b) 48 (c) 24 (d) 96

Ans: (b) 48

$$\begin{aligned} \text{Packets per carton} &= \frac{1152 \text{ packets}}{12 \text{ cartons}} = 96 \text{ packets/carton} \\ \text{Number of cartons} &= \frac{4608 \text{ packets}}{96 \text{ packets/carton}} = 48 \text{ cartons.} \end{aligned}$$

91. If meter cloth costs Rs.72, then what would be the cost of 4 meters of cloth?

- (a) Rs.369 (b) Rs.450 (c) Rs.396 (d) None of these

Ans: (d) None of these

$$4 \text{ meters} \times \text{Rs. } 72/\text{meter} = \text{Rs. } 288$$

92. 450 grams of cake costs 90 rupees. How much would a 2.5 kg cake cost?

- (a) Rs.450 (b) Rs.750 (c) Rs.400 (d) Rs.500

Ans: (d) Rs.500

$$\begin{aligned} \text{Price per gram: } &\text{Rs. } 90 / 450 \text{ grams} = \text{Rs. } 0.20 \text{ per gram.} & \boxed{2.5 \text{ kg} = 2500 \text{ grams}} \\ \text{Total cost: } &2500 \text{ grams} \times \text{Rs. } 0.20/\text{gram} = \text{Rs. } 500. \end{aligned}$$

93. If 16 men can finish a work in 72 days, then how many days 4 men would take to complete the task?

- (a) 288 (b) 298 (c) 285 (d) 256

Ans: (a) 288

Total work units: 16 men x 72 days = 1152 units of work
 Days for 4 men: 1152 units / 4 men = 288 days.

94. A map is given with a scale of 2 cm = 500 km. What is the distance between two places in km, if the distance in the map is 4.5 cm?

(a) 1225 km (b) 1325 km (c) 1125 km (d) 1425 km

Ans: (c) 1125 km

Distance per cm: 500 km / 2 cm = 250 km/cm.
 Actual distance: 4.5 cm x 250 km/cm = 1125 km

95. A typist types 80 pages in 2 hours 30 minutes. How long will he take to type 400 pages?

(a) 12.5 hours (b) 11.5 hours (c) 10.5 hours (d) 9.5 hours

Ans: (a) 12.5 hours

$$\begin{aligned} \text{Pages per hour} &= \frac{2 \text{ hours } 30 \text{ min}}{80} \\ 400 \text{ pages per hour} &= \frac{2 \text{ hours } 30 \text{ min}}{80} \times 400 = 12.5 \text{ hours} \end{aligned}$$

96. 45 horse can graze a field in 24 days. How many horses will graze the same field in 8 days?

(a) 125 (b) 135 (c) 145 (d) 150

Ans: (b) 135

$$\begin{aligned} N_1 \times D_1 &= N_2 \times D_2 \Rightarrow 45 \times 24 = 8 \times N_2 \\ N_2 &= \frac{45 \times 24}{8} = 135 \text{ horses} \end{aligned}$$

97. If 25 envelopes cost Rs.25, then what is the cost of 65 envelopes?

(a) Rs.64 (b) Rs.63 (c) Rs.65 (d) Rs.70

Ans: (c) Rs.65

98. 10 men can finish the construction of a boundary wall in 7 days. How many men need to be added to finish the work in half day?

(a) 80 (b) 140 (c) 50 (d) 130

Ans: (d) 130

total work is 10 men x 7 days = 70 man-days.
 number of men required is $\frac{70 \text{ man-days}}{0.5 \text{ days}} = 140 \text{ men}.$
 The number of men to be added is 140 men – 10 men = 130 men

99. In a map 1 cm represents 20 km. How much distance will be represented by 70.5 cm?

(a) 1410 km (b) 1400 km (c) 1420 km (d) None of these

Ans: (a) 1410 km

$$\text{The distance represented is } 70.5 \text{ cm} \times \frac{20 \text{ km}}{1 \text{ cm}} = 1410 \text{ km}$$

100. 12 meters of a steel pipe weighs 24 kg, then what would be the weight of 18 Meters pipe?

(a) 24 (b) 36 (c) 30 (d) 40

Ans: (b) 36

$$\text{The weight per meter is } \frac{24 \text{ kg}}{12 \text{ meters}} = 2 \text{ kg/meter.}$$

$$\text{The weight of 18 meters of pipe is } 18 \text{ meters} \times 2 \text{ kg/meter} = 36 \text{ kg}$$